

SUMANT KUMAR GUPTA | SR. AI/ML ENGINEER

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PROFESSIONAL SUMMARY:

Results-driven **Senior AI/ML Engineer** with **11 years** of experience in designing, developing, and delivering innovative software solutions. Demonstrates a deep understanding of web technologies and excels in high-paced, deadline-oriented environments. Recognized for adaptability, technical versatility, and a strong commitment to project success.

- **Comprehensive AI/ML Expertise:** Extensive experience in artificial intelligence (**AI**) and machine learning (**ML**) model development, deployment, and optimization. Proven track record of leveraging AI/ML techniques to drive data-driven decision-making and enhance business operations. Adept at using **TensorFlow, Scikit-learn, PyTorch**, and **Hugging Face Transformers** to build predictive models, deep learning solutions, and generative AI applications.
- **Large Language Models (LLMs):** Deep expertise in fine-tuning and integrating Large Language Models to enhance natural language understanding and generation.
- **NLP and GenAI Proficiency:** Skilled in Natural Language Processing (**NLP**) techniques for **text analysis, sentiment analysis**, and **chatbot development**. Experienced in using generative AI techniques to create synthetic data for model training and to develop advanced AI solutions.
- **Cloud and DevOps Acumen:** Extensive hands-on experience in the AWS cloud environment, including designing and implementing scalable solutions using **EC2, S3, Lambda, and RDS**. Expertise in containerization systems like **Docker** and container orchestration using **Kubernetes**. Proficient in configuring and managing **SOAP/RESTful** web services in JSON and XML formats.
- **Web Development and Big Data:** Comprehensive expertise in web development, specializing in Web Services and **Python**. Experienced with big data technologies like **Apache Spark** and **Hadoop** for large-scale data processing and analysis. Designed RESTful APIs using Node.js and Spring Boot for AI integrations. Built user-friendly dashboards and visualization interfaces with Angular and React to display model outcomes and insights.
- **Project Leadership and Collaboration:** Proven ability to lead cross-functional teams, manage projects, and deliver high-quality software solutions on time. I have a strong background in maintaining version control systems, including **SVN** and **GIT**. Well-versed in Shell Scripting (**BASH, SH**) in UNIX and Linux environments.

TECHNICAL SKILLS:

Programming Languages	Python, Java, javascript, SQL, Bash/Shell
Libraries/Frameworks	Scikit-learn, TensorFlow, PyTorch, Hugging Face Transformers, Keras, Pandas, NumPy, SciPy, Matplotlib, Seaborn, PyTables, NLTK, SpaCy, Transformers, OpenAI APIs, GPT, BERT, T5, LangChain, Synthetic data generation
Databases	MySQL, PostgreSQL, MS SQL Server, MongoDB
Cloud Technologies	Amazon Web Services (EC2, S3, Lambda, RDS, CloudWatch, Kinesis)
DevOps Tools	Docker, Kubernetes, Jenkins, GitHub Actions, MLflow (optional), Ansible, Chef, Splunk, ELK Stack
Visualization	MS Excel, Tableau
Web Development	HTML, CSS, JavaScript, Angular, React, Flask, Node.js, Spring Boot
Big Data Technologies	Apache Spark, Hadoop, Kafka
Versioning Tools	Git, GitHub, GitLab, TFS
API development	RESTful APIs, OpenAPI/Swagger
SDLC	Agile, Scrum, Waterfall, Jira, Confluence

EDUCATION:

Bachelor's Degree in Computer Science and Engineering, Anna University, Salem, TN, India

2014

PROFESSIONAL EXPERIENCE:

Senior AI/ML Engineer

Project: AI-Driven Patient Feedback Analyzer and Workflow Optimizer for Healthcare Platform

Built an AI-powered system to analyze patient feedback, predict service satisfaction, and optimize hospital workflows. The system integrated with EHRs and internal scheduling platforms to enhance both clinical and administrative efficiency.

Responsibilities:

- Designed and developed end-to-end machine learning pipelines using **Python, TensorFlow, Scikit-learn**, and **MLFlow**, supporting scalable training, evaluation, and deployment workflows.
- Built and deployed custom NLP models for sentiment and intent classification of patient reviews using techniques like **TF-IDF, Word2Vec**, and **BERT** embeddings.
- Implemented Spark-based **ETL** pipelines to efficiently handle and process large-scale, multi-source patient datasets from various clinical systems.
- Performed comprehensive text preprocessing using **NLTK and Scikit-learn**, including tokenization, stemming, and feature extraction to enhance NLP model performance.
- Engineered a reinforcement learning-based appointment scheduler that dynamically optimized slot allocation and reduced patient no-shows by 25%.
- Integrated predictive models with **HL7/FHIR APIs** to enable real-time synchronization with electronic health records (EHRs) and deliver actionable patient insights.
- Developed ML-powered anomaly detection alerts to identify critical feedback trends and trigger compliance escalation workflows.
- Designed and deployed **RESTful ML APIs** for real-time predictions, embedding AI functionality directly into hospital systems and patient-facing applications.
- Maintained strict **HIPAA** compliance by implementing secure deployment practices, data encryption, and role-based access control (RBAC).
- Created interactive Tableau dashboards to visualize sentiment analysis results, key model KPIs, and performance metrics for business stakeholders.
- Automated model packaging and deployment using Docker containers and Jenkins pipelines, streamlining CI/CD operations for machine learning models.
- Integrated **SonarQube** into CI/CD workflows to ensure code quality, maintainability, and adherence to security best practices.
- Managed model lifecycle and versioning with MLFlow, including experiment tracking, model registry, and automated retraining logic.
- Conducted **A/B testing** and real-world evaluations to validate model effectiveness and ensure robustness in production environments.
- Deployed scalable **ML services** on AWS, implementing monitoring, rollback, and alerting mechanisms to maintain service uptime and reliability.
- Collaborated with frontend developers to integrate real-time AI components into Angular 15 applications, enabling dynamic UI behavior based on live predictions.
- Led AI architecture design and planning within agile, cross-functional teams, aligning technical strategies with business goals.
- Documented **ML pipelines** and sprint progress using JIRA and Confluence, ensuring transparency and alignment across development teams.
- Collaborated closely with data engineering and DevOps teams to maintain robust, scalable, and fault-tolerant ML infrastructure.
- Implemented model explainability techniques such as **SHAP and LIME** to ensure transparency, fairness, and trust in AI-driven decisions.
- Achieved a **30%** boost in patient satisfaction scores, directly supporting critical business KPIs through data-driven enhancements in healthcare operations.

Environment: Python, TensorFlow, Scikit-learn, MLFlow, Spark, Hadoop, SQL, Docker, Jenkins, SonarQube, NLTK, BERT, Word2Vec, TF-IDF, Angular 15, RESTful APIs, Tableau, HL7/FHIR, Git, AWS, SHAP, LIME, JIRA, Confluence, Agile/Scrum, HIPAA Compliance

AI Engineer

Project: Smart Document Processing and Automated Insights (SDPAI)

Developed an AI-driven platform to process scanned documents, extract structured data, and generate automated analytics and insights for enterprise clients in banking and insurance sectors.

Responsibilities:

- Designed and implemented deep learning models for document classification, text extraction, and information retrieval using **Convolutional Neural Networks (CNNs)** and **Recurrent Neural Networks (RNNs)**.
- Integrated **Optical Character Recognition (OCR) tools (Tesseract, Azure Form Recognizer)** to extract text from scanned documents and images, improving data extraction accuracy by 30%.
- Built **Natural Language Processing (NLP)** pipelines to process raw text data, removing noise and preparing data for further analysis using libraries like spaCy and NLTK.
- Developed models for classifying documents into predefined categories such as invoices, contracts, and receipts, using supervised learning algorithms like **Random Forests and SVM**.
- Fine-tuned **pre-trained transformer models (BERT, LayoutLM)** to perform Named Entity Recognition (NER), extracting key information such as dates, customer names, and financial data.
- Utilized data augmentation techniques (synthetic data generation, noise addition) to expand training datasets, improving model robustness, especially with limited document data.
- Applied transfer learning to leverage pre-trained models for document analysis, reducing training time by **50%** and improving model accuracy for custom use cases.
- Deployed **AI models** and processing pipelines on AWS, ensuring scalability and performance through cloud-based infrastructure, using EC2, S3, Lambda, and Elastic Beanstalk.
- Utilized **Docker** for containerization and Kubernetes for orchestrating AI services, enabling seamless deployment and management of AI models across environments.
- Developed an automated data pipeline using **Python, Apache Airflow, and Apache Kafka** for ingesting, processing, and storing document data efficiently in real-time.
- Implemented continuous model monitoring to track **performance metrics (accuracy, precision, recall)** and fine-tuned models based on performance feedback.
- Built a system to generate automated reports summarizing key insights from processed documents, leveraging **NLP techniques** to extract trends and patterns.
- Integrated **AI models** with existing enterprise systems (CRM, ERP) using RESTful APIs, enabling seamless workflow automation and real-time document processing.
- Ensured compliance with data protection standards (**GDPR, HIPAA**) when handling sensitive customer information in the document processing system.
- Led model evaluation efforts by designing performance benchmarks, testing the models on real-world data, and optimizing hyperparameters using Grid Search and Random Search.
- Collaborated closely with UI/UX designers and backend engineers to design user-friendly interfaces for document processing and AI-driven insights visualization.
- Identified bottlenecks in processing workflows and optimized algorithms, leading to a **40%** reduction in processing time for large document batches.
- Built **data cleaning and preprocessing** scripts to handle noisy, inconsistent, and incomplete data, enhancing the quality and reliability of training datasets.
- Advocated for the use of ethical AI practices, ensuring that models were trained with diverse and balanced data to prevent bias and promote fairness in document processing.

Environment: Python, NLTK, spaCy, TensorFlow, PyTorch, Tesseract, Azure Form Recognizer, AWS (EC2, S3, Lambda), Docker, Kubernetes, Apache Kafka, Apache Airflow, scikit-learn (Random Forests, SVM), RESTful APIs (Flask, Django)

Python Developer**Responsibilities:**

- Spearheaded the development of an **AI-driven image mood detection app** using Python, OpenCV, and ML techniques during an internal hackathon, earning 2nd place recognition.

- Utilized exploratory data analysis (**EDA**) to uncover insights and trends in user engagement data.
- Built and evaluated machine learning models with scikit-learn to predict user behavior patterns.
- Created efficient **ETL** pipelines using Python and SQL for cleaning, transforming, and preparing datasets for ML training.
- Developed interactive data visualizations using **Matplotlib** and **Seaborn** to communicate model results and trends.
- Collaborate cross-functionally in an **Agile/Scrum** environment, contributing to sprint planning, daily stand-ups, and product reviews.
- Leveraged Postman for **API** testing and integration validation throughout the development cycle.
- Employed **Git** for version control and team collaboration, ensuring smooth branching and merging workflows.
- Worked in both **IntelliJ IDEA** and **VS Code**, adapting to project requirements and team preferences.
- Gained hands-on experience in combining software development with data science for building smart, user-centric features.

Environment: Python, OpenCV, Scikit-learn, Pandas, SQL, Matplotlib, Seaborn, Git, Agile/Scrum, Postman, IntelliJ, VS Code

Haptik, Mumbai, India

Sept 2014 – Nov 2015

Software Engineer

Responsibilities:

- Designed and developed conversational AI flows using natural language understanding (**NLU**) to enhance user interaction for chatbots across domains like e-commerce, travel, and customer support.
- Built intent classification and entity extraction models using traditional **ML algorithms** (Naive Bayes, SVM) and experimented with early deep learning techniques (RNN, LSTM) to improve bot comprehension.
- Created **data preprocessing** pipelines for large-scale conversational datasets, including tokenization, POS tagging, stemming/lemmatization, and feature extraction using TF-IDF and word embeddings.
- Implemented feedback loops to retrain and fine-tune models based on real-time user queries and interaction logs.
- Integrated **third-party APIs** to enable bots to perform transactions like booking, order tracking, FAQs, and appointment scheduling.
- Participated in **A/B testing** of dialog flows and response generation strategies to maximize user engagement and reduce fallback rates.
- Developed rule-based and hybrid systems combining deterministic **NLP techniques and ML models** for enhanced performance in low-resource settings.
- Collaborated with product managers and UX teams to iteratively design chatbot personality and tone for different use cases.
- Deployed **AI models** in production using containerization (Docker) and monitored performance using logs and analytics dashboards.

Environment: Python, scikit-learn, TensorFlow (early versions), Keras, NLTK, spaCy, Flask, RESTful APIs, MongoDB, Git, Jira, Linux, Docker